# EnvHarness: Awakening Static Worlds for Agent Learning

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HarnessRSI 63 ★★

evidence: retained

Paper

Code

## 1 What changes, what is kept

| | |
|---|---|
| **Stage** | Replay actions to set the start |
| **Contract** | Filter actions, responses, views |
| **Chain*** | Join tasks; both verifiers must pass |

**Observe**
5 unmodified rollouts per task
Read successes and failures

▼

**Diagnose + write**
Same-backbone designer model
Targets a flaw with a component set

▼

**Validate**
5 fresh rollouts; native verifier
Model judges SR, failures, timeouts

▼

**Keep / refine / reject**
Reject impossible or trivial tasks
At most 5 write–validate rounds

**Accepted wrapper → trajectories → skills**
Frozen policy + skills → original test tasks

*Chain is tested separately, not generated by the main EnvRigger loop. Backend and native scoring stay fixed; exposed responses may change.

**Acceptance is a model decision from training rollouts**—not a held-out gain check or an independent proof of solvability. Exhausted tasks may yield no component.

§§2–4; Appendix A; Table 8. Skills persist; main skill-learning model weights do not change.

## 2 Headline: +9.0 points is OOD

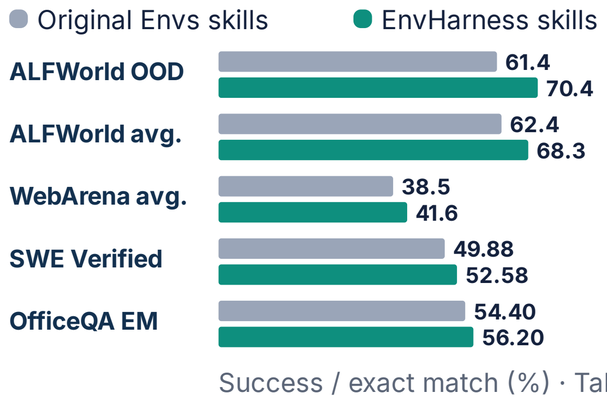



Table 2: three independent runs, mean ± SD. OOD: 61.4 ± 4.3% → 70.4 ± 2.3%. Table 3 supplies SDs, not an explicit repeat count. No CIs.

SWE steps: 55.01 → 49.61 (9.8% fewer vs Original Envs). SWE-smith SR: 50.12 ± 1.74%; EnvHarness: 52.58 ± 2.72%.

## 4 Where gains thin or reverse

- **WebArena GitLab:** EnvHarness 37.7 ± 3.1%, below VeriEnv 38.9 ± 5.6%. Shop Admin has the largest gain vs Original Envs: +6.2 points (Table 2).
- **Leave-one-type-out ALFWorld:** mean 60.6% → 63.7%; clean +16.4 points, heat −8.7, simple tied at 83.6%. Table 10 has no variance/CI.
- **RL OOD regression:** ALFWorld 89.6% → 88.8%; In-Dist 81.4% → 87.9%. Qwen3-8B-base, GRPO; seed 0, 150 epochs/steps, no variance/CI (Table 4, F.1).
- **Not always fewer steps:** Qwen3.6 27B 37.1 → 40.8; Claude Sonnet 4.6 25.4 → 25.6, vs original skills (Table 9).
- **Small margins:** SWE/Office mean gains overlap reported run spread; no significance test. Requires resettable textual environments.

## 3 Controls, splits and real cost

### Same count ≠ same interaction budget

Original Envs, GenEnv (ALFWorld), VeriEnv (WebArena), SWE-smith (SWE): same seed tasks, environment count, policy model, extraction/retrieval; common oracle verification access stated. Diversity is not quantified or matched.

Baselines do not receive EnvHarness's wrapper-writing / validation loop. No generation baseline for office tasks.

### Train-only generation; original-task tests

SWE: 100 Lite training tasks → 407 Verified issues not in Lite. OfficeQA: 50 train → 172 test. SpreadsheetBench: 100 train → 299 test tasks / 897 instances.

ALFWorld uses native seen/unseen splits; WebArena evaluates Reddit, Shopping, Shop Admin and GitLab. No semantic-duplicate or pretraining-contamination audit reported.

### Component control · Table 5

| Skill source | SR (%) | Steps |
|---|---|---|
| Original Envs | 49.88 | 55.01 |
| Stage/Contract | 52.58 | 49.61 |
| Chain only | 49.63 | 41.96 |
| Combined | **54.30** | 43.12 |

Test: single-environment SWE issues, not joined tasks. No variance/run count; no separate diagnosis or validation ablation.

### Generation tokens · estimated, Table 11

| Benchmark / method | Design | Total |
|---|---|---|
| ALFWorld / GenEnv | 38K | 64.2M |
| ALFWorld / EnvHarness | 1.46M | 228.0M |
| WebArena / VeriEnv | 20K | 137.8M |
| WebArena / EnvHarness | 1.58M | 137.3M |

Rollouts dominate; GenEnv simulates them, EnvHarness/VeriEnv execute them. One-time design cost per environment. No SWE-smith/Original Envs cost totals.

### Defaults and tuning are not the same

Shared defaults (Table 8); selection split and baseline tuning budgets unreported. Missing values include standard target SR band, skill-learning decoding temperature/step cap, RL learning rate.

## Takeaway

Target the learner's flaws by reshaping practice—not the task scorer. Retained skills help on original held-out tasks, but transfer is uneven and equal environment counts do not make generation compute equal.

Main backbones: Gemini-3.1-Flash-Lite (ALFWorld/WebArena); Gemini-3.5-Flash elsewhere. Designer = policy backbone. Sources: Tables 2–5, 7–11; §§3–5; Appendices A, E–H.

## Our read · HarnessRSI

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**Reports:** held-out tests, SDs, type-transfer, token costs, component control.

**Not reported:** universal compute parity, baseline tuning budgets, semantic leakage audit, diagnosis/gate isolation.

Full-paper additions to b23 noted in SOURCE.md; score unchanged.